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Inventor(s)	Ogawa; Takaya et al.

Drinking container holding device

Abstract

Provided is a drinking container holding device that is configured to heat the drinking container. The drinking container holding device includes a plastic holder configured to hold a drinking container on the inner side of the holder, a sheet-like heat transfer element located on the outer side of the holder and having a higher thermal conductivity than the holder, a sheet heating element located on the outer side of the heat transfer element and smaller than the heat transfer element, and an insulating element covering the sheet heating element on the outer side of the sheet heating element.

Inventors:	**Ogawa; Takaya (Yokosuka, JP), Saotome; Masataka (Yokosuka, JP), Toda; Masao (Kyoto, JP), Miyamoto; Kazuhiko (Kyoto, JP)**
Applicant:	**NIFCO INC.** (Yokosuka, JP); **KYOTECH CO., LTD.** (Kyoto, JP)
Family ID:	**86157898**
Assignee:	**NIFCO INC. (Yokosuka, JP); KYOTECH CO., LTD. (Kyoto, JP)**
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Primary Examiner: Ackun; Jacob K

Attorney, Agent or Firm: Tarolli, Sundheim, Covell & Tummino LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Stage Application filed under 35 U.S.C. § 371 of PCT/JP2022/037863, filed Oct. 11, 2022, and entitled "DRINKING CONTAINER HOLDING DEVICE", which claims priority from Japanese Patent Application No. 2021-174716, filed on Oct. 26, 2021. The entire contents of each of the above-identified patent applications are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a drinking container holding device.

BACKGROUND ART

(3) An example of a drinking container holder includes a holder body configured to hold a drinking container. The holder body includes an inner case and an outer case arranged outside the inner case. The inner case is formed from metal. The outer case is formed from a polyurethane resin. The inner case is in contact with a Peltier element (for example, refer to Patent Literature 1).

CITATION LIST

Patent Literature

(4) Patent Literature 1: Japanese Laid-Open Patent Publication No. 2016-068854

SUMMARY OF INVENTION

Technical Problem

(5) The Peltier element is used as a heater that heats a portion of the inner case, which is used as a thermal conductor. The inner case includes an inner surface of the holder body. When the drinking container has heated to a predetermined temperature, the temperature of the inner surface of the holder body, which can contact a hand of a user, may be overly increased at only a portion of the inner surface that is in contact with the heater.

Solution to Problem

(6) In an aspect of the present disclosure, a drinking container holding device is configured to heat a drinking container. The drinking container holding device includes a holder formed from resin and configured to hold the drinking container in the holder, a sheet-shaped thermal conductor arranged outside the holder and having a higher thermal conductivity than the holder, a sheet-like heating element arranged outside the thermal conductor, the sheet-like heating element being smaller than the thermal conductor, and a thermal insulator covering the sheet-like heating element outside the sheet-like heating element.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. **1** is an exploded perspective view showing a first embodiment of a drinking container holding device before being assembled.

(2) FIG. **2** is a perspective view of the drinking container holding device shown in FIG. **1** when assembled.

(3) FIG. **3** is a cross-sectional view showing the structure of the drinking container holding device shown in FIG. **1**.

(4) FIG. **4** is an enlarged cross-sectional view of region A shown in FIG. **3**.

(5) FIG. **5** is a side view showing the structure of an operating switch and a portion of a sheet-like heating element that are included in the drinking container holding device shown in FIG. **1**.

(6) FIG. **6** is a cross-sectional view showing the structure of the operating switch shown in FIG. **5** and a portion of the sheet-like heating element.

(7) FIG. **7** is a plan view showing the structure of a modified example of a thermal conducting sheet included in the drinking container holding device shown in FIG. **1**.

(8) FIG. **8** is a plan view showing the thermal conducting sheet shown in FIG. **7** and the structure

of a holder.
(9) FIG. **9** is a perspective view showing the structure of a thermal conductor of a drinking container holding device in a second embodiment.
(10) FIG. **10** is a perspective view showing the structure of a thermal conductor of a drinking container holding device in a third embodiment.
(11) FIG. **11** is a perspective view showing the structure of a thermal insulator of the drinking container holding device in the third embodiment.

DESCRIPTION OF EMBODIMENTS

First Embodiment

(12) A first embodiment of a drinking container holding device will now be described with reference to FIGS. **1** to **6**. FIG. **1** shows the drinking container holding device before being assembled.
(13) In FIG. **1**, a drinking container holding device **10** (hereafter, may be referred to as holding device) is configured to heat a drinking container. The drinking container may be, for example, a metal beverage can, a paper beverage cup, or a plastic beverage bottle. The holding device **10** includes a holder **11**, a thermal conductor **12**, a sheet-like heating element **13**, and a thermal insulator **14**. The holder **11** holds a drinking container in the holder **11**. The holder **11** is formed from resin.
(14) The thermal conductor **12** is arranged outside the holder **11**. The thermal conductor **12** is sheet-shaped. The thermal conductor **12** has a higher thermal conductivity than the holder **11**. The sheet-like heating element **13** is arranged outside the thermal conductor **12**. The sheet-like heating element **13** is smaller than the thermal conductor **12**. Hence, the sheet-like heating element **13** covers at least a portion of the thermal conductor **12**. The thermal insulator **14** is arranged outside the sheet-like heating element **13** and covers the sheet-like heating element **13**.
(15) In the holding device **10** of the present disclosure, heat is transferred from the sheet-like heating element **13** to the drinking container through the thermal conductor **12**, which is larger than the sheet-like heating element **13**, and the resin holder **11**. In this process, the thermal conductor **12** diffuses heat from the sheet-like heating element **13**, and the resin holder **11** diffuses heat from the thermal conductor **12**. The heat, which has diffused from the sheet-like heating element **13** through the thermal conductor **12** and the holder **11**, is transferred to the drinking container in a wide area. In addition, the resin holder **11** hampers direct transfer of heat from the drinking container to the thermal conductor **12** while diffusing heat of the drinking container in the resin holder **11**. Such heat diffusion by the resin holder **11** limits an uneven temperature distribution in the thermal conductor **12**. Thus, heat appropriately diffuses from the sheet-like heating element **13**. The diffusion of heat from the sheet-like heating element **13** and thermal insulation of the thermal insulator **14** result in uniform heating of the drinking container in a wide area while limiting an excessive increase in the temperature in a local area occupied by the sheet-like heating element **13**.
(16) The holder **11** is a resin mold component. The holder **11** may be formed from various synthetic resins. The holder **11** includes a tube **11**A and a flange **11**B. The tube **11**A includes two axial ends and a bottom wall closing one of the two axial ends of the tube **11**A. The tube **11**A includes an inner surface defining an accommodation cavity **11**AS. The accommodation cavity **11**AS is sized to accommodate various drinking containers. As described above, the drinking container may be, for example, a beverage can, a beverage cup, or a beverage bottle. The holder **11** may be shaped to accommodate only a specific drinking container. In the present embodiment, the tube **11**A is cylindrical. However, the tube **11**A may be truncated-cone-shaped or polygonal.
(17) The flange **11**B overhangs from an open end of the tube **11**A, which is one of the two ends, in a radial outward direction of the tube **11**A. The flange **11**B includes a through hole **11**BH extending through the flange **11**B in the thickness-wise direction. The flange **11**B includes a retainer **11**B**1** and a hook **11**B**2**. The flange **11**B includes one retainer **11**B**1**. The number of hooks **11**B**2** is the same as the number of clips **16**, which will be described later.

(18) The flange **11**B includes two opposing surfaces in the flange **11**B. One of the two opposing surfaces is an upper surface. The other surface is a lower surface. The retainer **11**B**1** projects downward from the lower surface of the flange **11**B. In a view facing the lower surface, the retainer **11**B**1** extends around the through hole **11**BH. Each hook **11**B**2** projects downward from the lower surface of the flange **11**B in the same manner as the retainer **11**B**1**.
(19) The thermal conductor **12** includes thermal conducting sheets **12**A. The thermal conducting sheets **12**A may be formed from various metals. In an example, the thermal conducting sheets **12**A are formed from aluminum. In the present embodiment, the thermal conductor **12** includes three thermal conducting sheets **12**A. The thermal conducting sheets **12**A are each rectangular. The thermal conducting sheets **12**A are curved along an outer surface of the tube **11**A. As described above, the tube **11**A is round. Therefore, in a cross section orthogonal to the axial direction of the tube **11**A, the thermal conducting sheets **12**A are each arcuate such that the center of curvature is located in the tube **11**A.
(20) In the axial direction of the tube **11**A, the thermal conducting sheet **12**A has a smaller width than the tube **11**A. The size, that is, the shape and the area, of the thermal conductor **12** is set so as to cover substantially the entirety of the lower half of the tube **11**A. The lower half of the tube **11**A is located below the retainer **11**B**1** and the hook **11**B**2** in the axial direction of the tube **11**A.
(21) The sheet-like heating element **13** is strip-shaped before being shaped in conformance with the outer shape of the tube **11**A. When mounted on the holding device **10**, the sheet-like heating element **13** includes an annular portion **13**A and a bent portion **13**B. The annular portion **13**A is shaped in accordance with the outer shape of the tube **11**A. The annular portion **13**A is shaped so that the thermal conducting sheets **12**A are sandwiched between the annular portion **13**A and the tube **11**A.
(22) The bent portion **13**B is joined to an end of the annular portion **13**A. The bent portion **13**B includes one or more bent portions. The bent portion **13**B is shaped so as to be connected to an operating switch **15**, which will be described later, and allow the operating switch **15** to be mounted on the flange **11**B.
(23) The sheet-like heating element **13** includes a resistive heating line **13**C and a sheet-like wire (refer to FIG. **6**). The resistive heating line **13**C extends on the annular portion **13**A in a sheet-like manner. The sheet-like wire is connected to the resistive heating line **13**C and a circuit substrate (refer to FIG. **6**), which will be described later, and supplies current to the resistive heating line **13**C. Thus, the sheet-like heating element **13** is configured to generate heat along the shape of the sheet in the entirety of the annular portion **13**A. In contrast, the bent portion **13**B includes only the sheet-like wire. The sheet-like heating element **13** includes an insulative cover **13**D. The cover **13**D covers the resistive heating line **13**C and the sheet-like wire.
(24) In the axial direction of the tube **11**A, the sheet-like heating element **13**, or the annular portion **13**A, has a smaller width than the thermal conductor **12**, or the thermal conducting sheet **12**A. Hence, the sheet-like heating element **13**, or the annular portion **13**A, is smaller than the thermal conductor **12** in size, in other words, in shape and area. Therefore, when the holding device **10** is assembled, the annular portion **13**A is located in the boundary of the thermal conductor **12**.
(25) The operating switch **15** is a mechanism for supplying current to the sheet-like heating element **13**. The operating switch **15** includes an operating button **15**A, a case **15**B, a packing **15**C, and a fastening member **15**D. The operating button **15**A is configured to control a start and a stop of the supply of current to the sheet-like heating element **13**. The case **15**B accommodates the circuit substrate, which will be described later. The circuit substrate is connected to the operating button **15**A and the sheet-like heating element **13**. When the operating switch **15** is mounted on the holder **11**, the packing **15**C is arranged between the operating button **15**A and the flange **11**B to restrict permeation of liquid from the flange **11**B toward the operating switch **15**. The fastening member **15**D fastens the case **15**B to the flange **11**B.
(26) The thermal insulator **14** is formed from resin. The thermal insulator **14** has the form of a tube

extending along the outer surface of the tube **11**A. The thermal insulator **14** includes two ends that are open in the axial direction of the tube **11**A. In the axial direction of the tube **11**A, the thermal insulator **14** has a greater width than the thermal conducting sheets **12**A and the annular portion **13**A. The size, that is, the shape and the area, of the thermal insulator **14** is set so as to cover the lower half of the tube **11**A.

(27) In the present embodiment, the thermal insulator **14** includes two ends in the circumferential direction of the thermal insulator **14**. The thermal insulator **14** is elastic so that the one end is separable from the other end in the circumferential direction of the thermal insulator **14**. The thermal insulator **14** may be formed from, for example, foam rubber or resin. Thus, when the thermal conductor **12** and the sheet-like heating element **13** are sandwiched between the thermal insulator **14** and the tube **11**A, the thermal insulator **14** is readily coupled to the tube **11**A.

(28) The holding device **10** includes clips **16**. The clips **16** are resin mold components. In the present embodiment, the holding device **10** includes four clips **16**. Alternatively, the holding device **10** may include three clips **16** or less and five clips **16** or more. The clips **16** are each hooked on a different one of the hooks **11**B**2**. The clips **16** extend in the axial direction of the tube **11**A. The clips **16** are configured to fix the holding device **10** to a subject in which the holding device **10** is installed. The installation subject of the holding device **10** is, for example, a vehicle.

(29) FIG. **2** shows the assembled holding device **10**.

(30) As shown in FIG. **2**, when the holding device **10** is assembled, the thermal insulator **14** covers the lower half of the tube **11**A so that the thermal conducting sheets **12**A and the annular portion **13**A are sandwiched between the thermal insulator **14** and the tube **11**A. The clips **16**, which are hooked on the hooks **11**B**2**, and the case **15**B, which is retained by the retainer **11**B**1**, are located at an outer side of the thermal insulator **14** in the radial direction of the tube **11**A and above the thermal insulator **14** in the axial direction of the tube **11**A. The bent portion **13**B of the sheet-like heating element **13** is exposed from the thermal insulator **14**.

(31) The operating switch **15** is retained by the retainer **11**B**1** so that the top surface of the operating button **15**A is exposed from the through hole **11**BH in a view facing the upper surface of the flange **11**B. Thus, the operating switch **15** is mounted on the holder **11**. As described above, since the operating switch **15** is mounted on the holder **11**, the installation subject of the holding device **10** does not need a separate structure for mounting the operating switch **15** in addition to the holding device **10**. The operating button **15**A, located in the through hole **11**BH of the flange **11**B, is pressed by a user of the holding device **10**.

(32) FIG. **3** is a cross-sectional view showing the structure of the holding device **10**.

(33) As shown in FIG. **3** and described above, the holder **11** includes the tube **11**A. The tube **11**A includes a circumferential wall **11**A**1** and a bottom wall **11**A**2**. The circumferential wall **11**A**1** has the form of a round tube surrounding a drinking container. The bottom wall **11**A**2** supports the drinking container in the holder **11**. The thermal insulator **14** has the form of a round tube and covers the lower half of the circumferential wall **11**A**1**.

(34) FIG. **4** is an enlarged view of the structure in region A shown in FIG. **3**.

(35) As shown in FIG. **4**, the circumferential wall **11**A**1** includes an outer surface that is in contact with the thermal conducting sheet **12**A of the thermal conductor **12**. Thus, the holding device **10** surrounds the drinking container and heats the drinking container. The entirety of the drinking container is heated in a preferred manner.

(36) The outer surface of the thermal conducting sheets **12**A is in contact with the annular portion **13**A of the sheet-like heating element **13**. Thus, when the sheet-like heating element **13** generates heat, the heat is transferred to the thermal conductor **12** more efficiently than in a structure having a gap between the annular portion **13**A and the thermal conducting sheets **12**A.

(37) The thermal insulator **14** may include an inner surface having rigidity that allows the inner surface to be deformed in conformance with the outer surface of the thermal conducting sheets **12**A and the outer surface of the annular portion **13**A. As described above, the thermal insulator **14** may

be formed from, for example, foam rubber or resin to have a rigidity that allows the inner surface of the thermal insulator **14** to be deformed in conformance with the outer surface of the thermal conducting sheets **12**A and the outer surface of the annular portion **13**A. Thus, a gap is not formed between thermal insulator **14** and the thermal conducting sheets **12**A. This further limits discharging of heat from the thermal conductor **12** to the exterior.
(38) The structure of the operating switch **15** will now be described in more detail with reference to FIGS. **5** and **6**. FIG. **5** is a side view showing the structure of the operating switch **15**. FIG. **6** is a cross-sectional view showing the structure of the operating switch **15** taken in the vertical direction.
(39) As shown in FIG. **5** and described above, the operating switch **15** includes the operating button **15**A and the case **15**B. The bent portion **13**B of the sheet-like heating element **13** includes one end connected to the annular portion **13**A and an opposite end located in the case **15**B.
(40) As shown in FIG. **6**, the case **15**B of the operating switch **15** accommodates a circuit substrate **15**E. The case **15**B includes an upper case **15**B**1** and a lower case **15**B**2**. The operating button **15**A is located on the upper case **15**B**1**. The circuit substrate **15**E is located on an upper portion of the lower case **15**B**2** and is covered by the upper case **15**B**1**. The operating switch **15** has the circuit substrate **15**E switch between supply and interruption of current. The operating switch **15** is configured to have the circuit substrate **15**E switch between supply and interruption of current when the operating button **15**A is pressed.
(41) The circuit substrate **15**E supplies current to the sheet-like heating element **13**. As described above, the sheet-like heating element **13** includes the resistive heating line **13**C (refer to FIG. **1**) and the sheet-like wire **13**E. The sheet-like wire **13**E includes a first end connected to the resistive heating line **13**C and a second end connected to the circuit substrate **15**E defining external connection terminals **13**E**1**. The external connection terminals **13**E**1** of the sheet-like wire **13**E are exposed from the cover **13**D.
(42) The circuit substrate **15**E includes fitting holes **15**EH into which the external connection terminals **13**E**1** of the sheet-like wire **13**E are fitted. The fitting holes **15**EH extend through the circuit substrate **15**E in a thickness-wise direction of the circuit substrate **15**E. The circuit substrate **15**E includes two fitting holes **15**EH. When fitted into the fitting holes **15**EH, the external connection terminals **13**E**1** are soldered. More specifically, solder **15**E**1** is applied to the circuit substrate **15**E around the fitting holes **15**EH. The solder **15**E**1** covers the fitting holes **15**EH and the portions of the external connection terminals **13**E**1** fitted into the fitting holes **15**EH.
(43) This structure eliminates the need for a further member between the sheet-like heating element **13** and the circuit substrate **15**E, thereby reducing the cost for manufacturing the holding device **10**. The further member is, for example, a cord or a connector.
(44) As described above, the drinking container holding device according to the first embodiment has the following advantages.
(45) (1-1) While the drinking container is uniformly heated in a wide area, the temperature is not overly increased in a local area occupied by the sheet-like heating element **13**.
(46) (1-2) The holding device **10** surrounds the drinking container and heats the drinking container. Thus, the entirety of the drinking container is heated in a preferred manner.
(47) (1-3) There is no need for a further member between the sheet-like heating element **13** and the circuit substrate **15**E. This reduces the cost for manufacturing the holding device **10**.
(48) (1-4) Since the operating switch **15** is mounted on the holder **11**, the installation subject of the holding device **10** does not need a separate structure for mounting the operating switch **15** in addition to the holding device **10**.
(49) (1-5) The thermal conducting sheets **12**A each have a smaller area as compared to a structure in which the thermal conductor **12** includes only a single thermal conducting sheet. Thus, the thermal conducting sheets **12**A are easily manipulated.
(50) The embodiment may be modified as follows.
(51) [Operating Switch]

(52) The operating switch **15** does not necessarily have to be mounted on the holder **11**. In this case, for example, the operating switch **15** may be mounted on the installation subject of the holding device **10** at a position separated from the holder **11**.
(53) [Sheet-Like Heating Element]
(54) The external connection terminals **13**E**1** of the sheet-like heating element **13** do not necessarily have to be soldered to the circuit substrate **15**E. In this case, for example, wires may be arranged between the circuit substrate **15**E and the external connection terminals **13**E**1** of the sheet-like heating element **13** to connect the external connection terminals **13**E**1** to the circuit substrate **15**E.
(55) [Thermal Conductor]
(56) The thermal conductor **12** may include a thermal conducting sheet, which will be described with reference to FIGS. **7** and **8**.
(57) As shown in FIG. **7**, each thermal conducting sheet **22**A is hexagonal. The thermal conducting sheet **22**A has the form of a hexagon obtained by cutting off two corners of a rectangle located next to each other in a longitudinal direction. The thermal conducting sheet **22**A includes two ends **22**A**1** in the longitudinal direction.
(58) As shown in FIG. **8**, a circumferential wall **21**A**1** has a tubular surface shaped as a truncated cone. The thermal conducting sheets **22**A of the thermal conductor **12** are attached to an outer surface of the circumferential wall **21**A**1**. Each thermal conducting sheet **22**A is attached to the outer surface so that one side of the thermal conducting sheet **22**A extends in a circumferential direction of the outer surface. The thermal conducting sheet **22**A includes ends **22**A**1**, each of which overlaps an end of an adjacent one of the thermal conducting sheets **22**A.
(59) The thermal conductor **12** described above has the following advantages.
(60) (1-6) The thermal conductor is arranged on the truncated-cone-shaped circumferential wall **21**A**1** so that the thermal conducting sheets **22**A are aligned with each other at the height-wise position and surround the circumferential wall **21**A**1**. Thus, even when the holder **11** has a tubular surface shaped as a truncated cone, a desired range of the drinking container is heated in a preferred manner.
(61) The hexagonal thermal conducting sheet **22**A has the following advantages over a rectangular thermal conducting sheet **22**A. When a rectangular thermal conducting sheet is wrapped around along an outer surface of the truncated-cone-shaped circumferential wall **21**A**1**, longitudinal ends of the thermal conducting sheet will be located above a longitudinal center of the thermal conducting sheet in the axial direction of the circumferential wall **21**A**1**. In this regard, if the longitudinal center of the thermal conducting sheet is increased in width, the longitudinal ends may exceed a cover region that needs to be covered by the thermal conducting sheet in the axial direction of the circumferential wall **21**A**1**. Hence, to maintain the rectangular shape of the thermal conducting sheet, the thermal conducting sheet needs to be decreased in width so that the longitudinal ends will not exceed the cover region.
(62) In the hexagonal thermal conducting sheet **22**A, the corners of the longitudinal ends are cut off. This allows the hexagonal thermal conducting sheet **22**A to have an increased width with which the longitudinal ends of the rectangular thermal conducting sheet **22**A may exceed the cover region. In other words, even when the thermal conducting sheet **22**A is increased in width, the longitudinal ends are not likely to exceed the cover region as compared to a rectangular thermal conducting sheet. As a result, the range in which the thermal conductor **12** transfers heat from the sheet-like heating element **13** may extend.
(63) The thermal conductor **12** may include two thermal conducting sheets **12**A or less and four thermal conducting sheets **12**A or more. When the thermal conductor **12** includes two thermal conducting sheets **12**A or more, the thermal conducting sheets **12**A may be mounted on the holding device **10** so that two thermal conducting sheets **12**A or more are located next to each other in the axial direction of the tube **11**A.
(64) The thermal conductor **12** may be arranged outside the holder **11** at the outside of the bottom

wall **11**A**2** without being arranged at the outside of the circumferential wall **11**A**1**. In this case, the sheet-like heating element **13** and the thermal insulator **14** may also be located at only the outside of the bottom wall **11**A**2**. This structure still obtains an advantage according to the above advantage (1-1).

Second Embodiment

(65) A second embodiment of a drinking container holding device will now be described with reference to FIG. **9**. The second embodiment differs from the first embodiment in the shape of a thermal conducting sheet of a thermal conductor. Thus, the differences will be described in detail. Components of the drinking holding device other than the thermal conductor will not be described in detail.

(66) FIG. **9** is a perspective view showing the structure of a thermal conductor.

(67) As shown in FIG. **9**, the thermal conductor includes a single thermal conducting sheet **32**. The thermal conducting sheet **32** is strip-shaped and is wrapped around an outer surface of the circumferential wall **11**A**1** in the circumferential direction of the outer surface. The thermal conducting sheet **32** includes slits **32**S. Each of the slits **32**S extends from one side of the outer surface extending in the circumferential direction in a direction intersecting the circumferential direction of the outer surface. The slits **32**S are arranged at intervals in the circumferential direction of the outer surface.

(68) In the example shown in FIG. **9**, the thermal conducting sheet **32** is sized to cover the lower half of the circumferential wall **11**A**1** in the same manner as the thermal conducting sheets **12**A of the first embodiment. The thermal conducting sheet **32** includes two opposing sides in the axial direction of the circumferential wall **11**A**1**. One of the two sides is an upper side, and the other is a lower side. The slits **32**S extend from the lower side toward the upper side in the axial direction of the circumferential wall **11**A**1**. The slits **32**S are arranged at equal intervals in the circumferential direction.

(69) Thus, when the thermal conducting sheet **32** is wrapped around the outer surface of the circumferential wall **11**A**1**, the width of the slits **32**S may be changed in the circumferential direction of the circumferential wall **11**A**1** as compared to before the thermal conducting sheet **32** is wrapped. In addition, when the thermal conducting sheet **32** is wrapped around the outer surface of the circumferential wall **11**A**1**, the portion located between adjacent ones of the slits **32**S is deformable. Thus, the thermal conducting sheet **32** including the slits **32**S is readily attached in conformance with the structure of the holder **11**.

(70) Each slit **32**S includes an end located at the lower side, defining a basal end, and an end located between the lower side and the upper side, defining a distal end. The thermal conducting sheet **32** includes a through hole **32**H continuous with the distal end of the slit **32**S. The through holes **32**H extend through the thermal conducting sheet **32** in the thickness-wise direction of the thermal conducting sheet **32**. In a view facing the outer surface of the thermal conducting sheet **32**, the contour of the through hole **32**H is circular.

(71) As described above, the thermal conducting sheet **32** includes the through holes **32**H continuous with the distal ends of the slits **32**S. Thus, when the portions of the thermal conducting sheet **32** located between adjacent ones of the slits **32**S are deformed in the circumferential direction of the circumferential wall **11**A**1** or are bent in the radial direction of the circumferential wall **11**A**1**, the thermal conducting sheet **32** is less likely to be torn at the distal ends of the slits **32**S.

(72) As described above, the drinking container holding device according to the second embodiment has the following advantage in addition to the above advantages (1-1) to (1-4).

(73) (2-1) The thermal conducting sheet **32** is readily attached in conformance with the structure of the holder **11**.

(74) The second embodiment may be modified as follows.

(75) [Thermal Conducting Sheet]

(76) The through holes **32**H may be omitted from the thermal conducting sheet **32**. Even in this case, the thermal conducting sheet **32** including the slits **32**S obtains an advantage according to the above advantage (2-1).
(77) The direction in which the slits **32**S extend is not limited to the axial direction of the circumferential wall **11**A**1** and may be a direction extending from the lower side toward the upper side and intersecting with the axial direction. Even in this case, an advantage according to the above advantage (2-1) is obtained.
(78) The slits **32**S may extend from the upper side toward the lower side of the thermal conducting sheet **32**. More specifically, the basal end of the slit **32**S may be located at the upper side, and the distal end may be located between the upper side and the lower side. In this case, following advantages are obtained.
(79) (2-2) When the circumferential wall **11**A**1** has the form of a truncated cone extending in the axial direction, for example, the width of the slit **32**S in the circumferential direction of the circumferential wall **11**A**1** may be gradually increased in the axial direction, that is, the direction extending from the lower side toward the upper side of the thermal conducting sheet **32**. Thus, the single thermal conducting sheet **32** is readily attached in conformance with the outer surface of the circumferential wall **11**A**1**.
(80) The width of the thermal conducting sheet **32** may be greater than the width of the lower half of the circumferential wall **11**A**1**. In this case, the distal end of the slit **32**S may be located above the bottom wall **11**A**2**. This structure obtains the following advantages.
(81) (2-3) The portions of the thermal conducting sheet **32** including the slits **32**S may extend beyond the circumferential wall **11**A**1** toward the bottom wall **11**A**2**. The portions extending to the bottom surface of the bottom wall **11**A**2** may be attached to the bottom surface so that the extended portions are bent. The thermal conducting sheet **32** transfers heat generated by the sheet-like heating element **13** along the bottom wall **11**A**2**.
(82) When the circumferential wall **11**A**1** has the form of a truncated cone extending in the axial direction and the slits **32**S are arranged at equal intervals, the portions of the thermal conducting sheet **32** covering a lower portion of the circumferential wall **11**A**1** overlap one another, and the portions of the thermal conducting sheet **32** covering the bottom wall **11**A**2** overlap one another. In particular, in the portions of the thermal conducting sheet **32** covering the bottom wall **11**A**2**, the overlap is increased in size, and thus the volume of the thermal conducting sheet **32** per unit area is increased. Hence, even when the sheet-like heating element **13** is located on only the circumferential wall **11**A**1**, heat generated by the sheet-like heating element **13** is readily transferred to the bottom wall **11**A**2**. As a result, the entirety of the holder **11** is readily heated.
(83) Other Modified Examples
(84) The drinking holding device of the second embodiment may be implemented by combining the modified example of the first embodiment related to the operating switch **15** with the modified example of the first embodiment related to the sheet-like heating element **13**.
Third Embodiment
(85) A third embodiment of a drinking container holding device will now be described with reference to FIGS. **10** and **11**. The third embodiment differs from the first embodiment in the shape of a thermal conducting sheet of a thermal conductor and the structure of a thermal insulator. Thus, the differences will be described in detail. Components of the drinking holding device other than the thermal conductor and the thermal insulator will not be described in detail.
(86) FIG. **10** is a perspective view showing the structure of a thermal conductor.
(87) As shown in FIG. **10**, a thermal conductor **42** includes a circumferential wall sheet **42**A and a bottom wall sheet **42**B. The circumferential wall sheet **42**A has the form of a round tube extending along the outer surface of the circumferential wall **11**A**1**. The circumferential wall sheet **42**A is a single strip-shaped sheet extending along the outer surface of the circumferential wall **11**A**1**.
(88) The bottom wall sheet **42**B includes a bottom wall portion **42**B**1** and a side wall portion **42**B**2**.

The bottom wall portion **42**B**1** is circular and extends along the bottom wall **11**A**2**. The side wall portion **42**B**2** is annular and extends upward from the edge of the bottom wall portion **42**B**1**. The side wall portion **42**B**2** includes slits **42**BS. Each slit **42**BS includes a distal end at the edge of the bottom wall portion **42**B**1** and extends through the side wall portion **42**B**2** in the axial direction of the circumferential wall **11**A**1**. The slits **42**BS are arranged at equal intervals in the circumferential direction of the circumferential wall **11**A**1**.
(89) The edge of the bottom wall portion **42**B**1** includes through holes **42**BH. The through holes **42**BH extend through the bottom wall portion **42**B**1** in the thickness-wise direction of the bottom wall portion **42**B**1**. The through holes **42**BH are arranged at equal intervals in the circumferential direction of the circumferential wall **11**A**1**. The through holes **42**BH are each continuous with a different one of the slits **42**BS. Thus, when the side wall portion **42**B**2** is bent, the bottom wall sheet **42**B is less likely to be torn at the distal ends of the slits **42**BS.
(90) When the thermal conductor **42** is coupled to the holding device **10**, the side wall portion **42**B**2** of the bottom wall sheet **42**B covers the lower end of the circumferential wall sheet **42**A. Thus, heat generated by the sheet-like heating element **13** is transferred to the bottom wall sheet **42**B through the circumferential wall sheet **42**A. When the thermal conductor **42** is coupled to the holding device **10**, the bottom wall portion **42**B**1** of the bottom wall sheet **42**B is in contact with the bottom surface of the bottom wall **11**A**2**. The drinking container continuously contacts the bottom wall **11**A**2**, which is the heat source, because of the weight of the drinking container. Thus, the drinking container is heated in a preferred manner.
(91) FIG. **11** is a perspective view showing the structure of a thermal insulator.
(92) As shown in FIG. **11**, a thermal insulator **44** includes a circumferential wall portion **44**A and a bottom wall portion **44**B. The circumferential wall portion **44**A has the form of a round tube extending along the outer surface of the circumferential wall **11**A**1**. The bottom wall portion **44**B is disc-shaped in conformance with the shape of the bottom wall **11**A**2**. When the thermal insulator **44** is coupled to the holding device **10**, the circumferential wall portion **44**A covers the outer surface of the circumferential wall sheet **42**A, and the bottom wall portion **44**B covers the bottom surface of the bottom wall sheet **42**B.
(93) As described above, the drinking container holding device according to the third embodiment has the following advantage in addition to the advantages (1-1) to (1-4).
(94) (3-1) The drinking container continuously contacts the heat source because of the weight of the drinking container. Thus, the drinking container is heated in a preferred manner.
(95) The third embodiment may be modified as follows.
(96) [Thermal Conducting Sheet]
(97) The through holes **42**BH may be omitted from the bottom wall sheet **42**B. Even in this case, the thermal conductor **42** including the bottom wall sheet **42**B obtains an advantage according to the advantage (3-1) described above.
(98) The slits **42**BS may be omitted from the bottom wall sheet **42**B. Even in this case, the bottom wall sheet **42**B that includes the bottom wall portion **42**B**1** and the side wall portion **42**B**2** may be formed by, for example, forming gathers in the boundary between the bottom wall portion **42**B**1** and the side wall portion **42**B**2** when bending the side wall portion **42**B**2** upward from the bottom wall portion **42**B**1**. Even in this case, the thermal conductor **42** including the bottom wall sheet **42**B obtains an advantage according to the advantage (3-1) described above.
(99) [Sheet-Like Heating Element]
(100) The sheet-like heating element may be arranged outside the bottom wall sheet **42**B in addition to the outside of the circumferential wall sheet **42**A. In this case, the drinking container held by the holding device **10** is further readily heated.
(101) The sheet-like heating element may be arranged at the outside of the bottom wall sheet **42**B without being arranged at the outside of the circumferential wall sheet **42**A. Even in this case, heat generated by the sheet-like heating element transfers to the circumferential wall sheet **42**A through

the bottom wall sheet **42**B. Thus, the entirety of the dinking container is readily heated.
(102) Other Modified Examples
(103) The drinking holding device of the third embodiment may be implemented by combining the modified example of the first embodiment related to the operating switch **15** with the modified example of the first embodiment related to the sheet-like heating element **13**. The thermal insulator **44** of the drinking holding device of the third embodiment may be combined with the drinking holding device of the first embodiment.
(104) The drinking holding device of the third embodiment may be implemented in combination with the drinking holding device of the second embodiment. More specifically, in the drinking holding device of the second embodiment, the thermal conductor may further include the bottom wall sheet **42**B of the thermal conductor **42** of the third embodiment in addition to the thermal conducting sheet **32**. The drinking holding device of the second embodiment may include the thermal insulator **44** of the drinking holding device of the third embodiment.
(105) [Common Challenge]
(106) The drinking container holding device of the present disclosure takes on a common challenge for increasing convenience in a vehicle cabin when the drinking container holding device is used in a vehicle. In particular, the drinking holding device solves a proposition for increasing the usability of a drinking container holding device.

Claims

1. A drinking container holding device configured to heat a drinking container, the drinking container holding device, comprising: a holder formed from resin and configured to hold the drinking container in the holder; a sheet-shaped thermal conductor arranged outside the holder and having a higher thermal conductivity than the holder; a sheet-like heating element arranged outside the thermal conductor, the sheet-like heating element being smaller than the thermal conductor; and a thermal insulator covering the sheet-like heating element outside the sheet-like heating element.
2. The drinking container holding device according to claim 1, wherein the holder includes a round, tubular circumferential wall configured to surround the drinking container, and the circumferential wall includes an outer surface in contact with the thermal conductor.
3. The drinking container holding device according to claim 2, wherein the thermal conductor includes thermal conducting sheets.
4. The drinking container holding device according to claim 3, wherein the circumferential wall is truncated-cone-shaped, and each of the thermal conducting sheets is hexagonal and is attached to the outer surface of the circumferential wall so that one side of each of the thermal conducting sheets extends in a circumferential direction of the outer surface of the circumferential wall.
5. The drinking container holding device according to claim 2, wherein the thermal conductor includes a single strip-shaped thermal conducting sheet wrapped around the outer surface of the circumferential wall in a circumferential direction of the outer surface, the thermal conducting sheet includes slits extending in a direction intersecting the circumferential direction from one side that extends in the circumferential direction, and the slits are arranged at an interval in the circumferential direction.
6. The drinking container holding device according to claim 1, wherein the holder includes a bottom wall configured to support the drinking container, and the bottom wall includes a bottom surface in contact with the thermal conductor.
7. The drinking container holding device according to claim 1, further comprising: a circuit substrate configured to supply current to the sheet-like heating element, wherein the sheet-like heating element includes a resistive heating line extending in a sheet-like manner, and a sheet-like wire including a first end connected to the resistive heating line and a second end connected to the circuit substrate, the second end defining an external connection terminal, the circuit substrate

includes a fitting hole into which the external connection terminal of the sheet-like wire is fitted, and the external connection terminal is fitted into the fitting hole and soldered.

8. The drinking container holding device according to claim 7, further comprising: an operating switch accommodating the circuit substrate and being configured to switch between supply and interruption of current to the circuit substrate, wherein the operating switch is mounted on the holder.